Insertion Sort Algorithm

The Insertion Sort Algorithm

An insertion sort is also quite simple in the way it works but is more efficient than a bubble sort. It steps through a list of data and for each item it finds it compares itself with the previous items of data:

If it is greater than the previous item, it is inserted back into its same position.

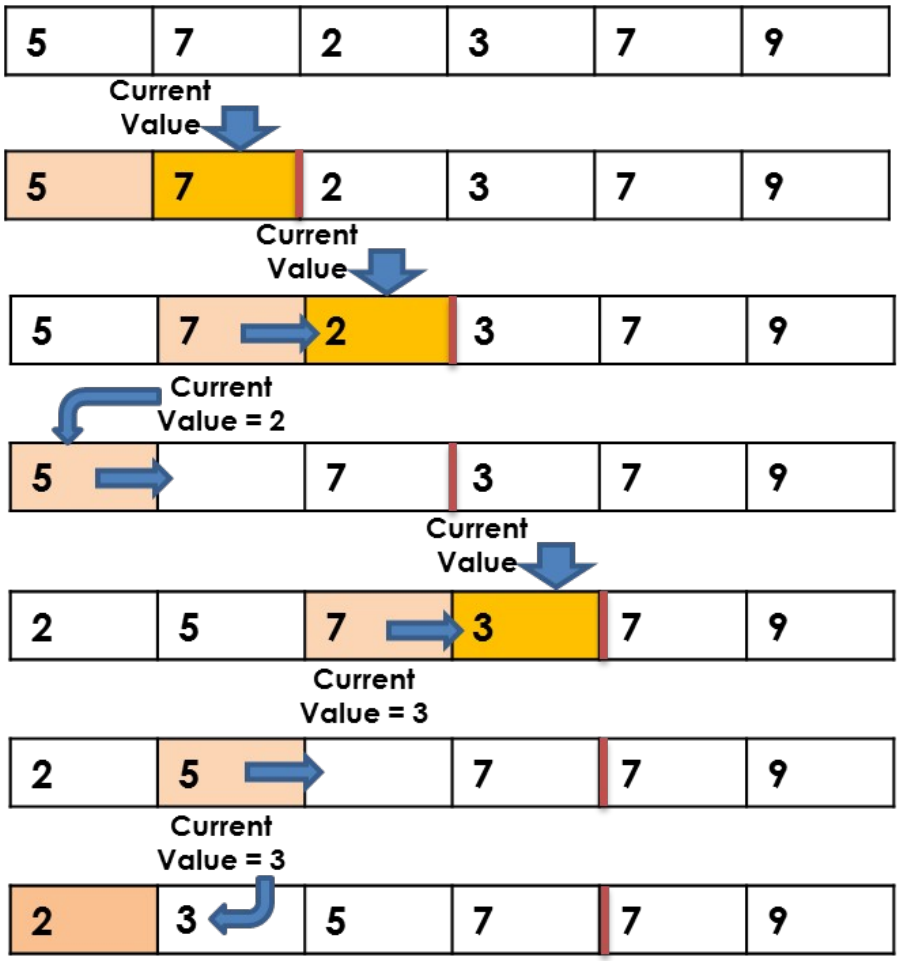
If it is smaller than the previous, it is then compared with the item before that.

...if it is greater than the item 2 positions back, it is inserted in front of it.

...else, if it is smaller than the item 2 positions back, it is compared with the item before it

...and so on and so forth, until it finds it is correct position.

An Example:



The 2nd item is compared with the 1st. As 5 isn't greater than 7, no change occurs.

Now 2 is the current value. It is stored in a temporary variable. As 7 is greater than 2, 7 overwrites 2 in the list.

Now 5 is compared with the current value. 5 is greater so it is copied into the next list position. 2 is inserted back in the list in the correct position.

Now 3 is the current value. It is stored in a temporary variable. As 7 is greater than 3, 7 overwrites 3 in the list.

Now 5 is compared with the current value. 5 is greater so it is copied into the next list position.

Insertion

Now 2 is compared with the current value. 2 is not greater so 3 is inserted back in the list in the correct position.

Sort Algorithm Pseudocode Example:

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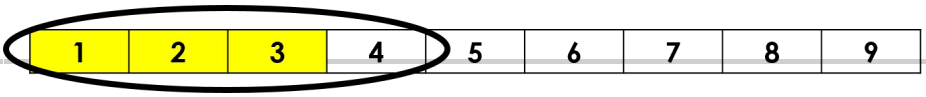
```
FOR position 1 to lengthOfdataset:
        currentValue = dataset[position]
        pos = position

        WHILE pos > 0 AND dataset[pos-1] > currentValue:
                dataset[pos] = dataset[pos-1]
                pos = pos – 1

        dataset[pos] = currentValue
```

Questions *(The question zone you choose must either match your target grade or be higher!)*

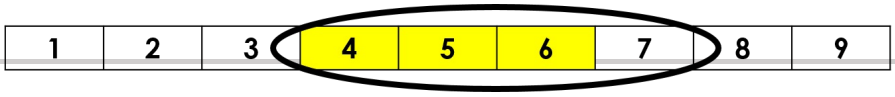
Question Zone 1-3



1. What is the purpose of the 'Insertion Sort Algorithm'?. [1]
2. What final steps would an insertion sort algorithm carry out in order to sort this semi-sorted dataset? [3]

   [5, 10, 15, 20, 0]
3. Describe how the 'Insertion Sort' works. [3]

Question Zone 4-6

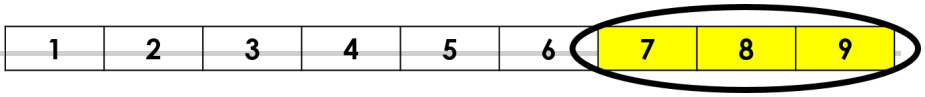


1. Describe how the 'Insertion Sort' works. [3]
2. How many comparisons and swaps would an insertion sort algorithm carry out in order to sort this semi-sorted dataset? [3]

   [5, 10, 15, 20, 0]
3. Show the steps that will take place to sort the following dataset, during an 'Insertion Sort'. [5]

   [10, 5, 20, 15, 0]

Questions Zone 7-9



1. Show the steps that will take place to sort the following dataset, during an 'Insertion Sort'. [5]

   [10, 5, 20, 15, 0]
2. How many comparisons and swaps are required to sort the following set of data using an 'Insertion Sort'? You must show your working out. [4]

   [2, 5, 4, 8, 7, 6]
3. What is the maximum number of comparisons and swaps that an 'Insertion Sort Algorithm' could make if there are 4 items of data in a dataset? You must show your working out. [5]

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Insertion Sort: *A method of sorting a set of data by repeatedly comparing data items with previous items in a dataset, before inserting them into the correct position.*

Comparison: *During the insertion sorting process, the data item being sorted at that time, will be* ***compared*** *with each previous data item.*

Algorithm*: A set of steps/instructions needed to complete a software task.*

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Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy, as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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_____

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